

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Order Instituting Rulemaking to
Modernize the Electric Grid for a High
Distributed Energy Resources Future.

Rulemaking 21-06-017
(Filed June 24, 2021)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE
ADMINISTRATIVE LAW JUDGE’S RULING PROVIDING TWO WORKING GROUP
REPORTS AND DIRECTING RESPONSES TO QUESTIONS ON REPORTS**

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments on the *Administrative Law Judge’s Ruling Providing Two Working Group Reports and Directing Responses to Questions on Reports* (“Ruling”), filed May 29, 2024, as modified by the *Administrative Law Judges’ Ruling Providing Uncorrupted Attachments*, filed June 3, 2024, and the *Email Ruling Partially Granting Extension of Time for Comments*, filed June 11, 2024.

I. INTRODUCTION.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of EVs and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC appreciates the work of all the stakeholders in the Smart Inverter Operationalization Working Group (“SIOWG”). These reports provide stakeholders with a comprehensive view of the working groups and many important takeaways for the Commission to consider in the next stage of this proceeding and beyond. California is in the middle of several paradigm shifts related to our electric system, with the interconnection of many distributed energy resources (“DERs”) being one of them. While DERs have traditionally been thought of as customer generation systems, dynamic loads are also important DERs that can be leveraged for grid benefits.

There are also resources that can provide both dynamic load and exports, such as electric vehicles (“EVs”) with bidirectional charging capabilities (“V2X”). However, overall, the distribution system does not have sufficient capacity to accommodate current requests to connect EV charging load and given regulatory requirements placed on automakers and fleet operators from the California Air Resources Board (“CARB”), EV charging load is only expected to increase. To accommodate this load, the Commission must change many aspects of current grid planning and energization processes.

Updating energization, interconnection, and grid planning practices to allow flexible DERs, both load and generation, to connect to the grid is critical not only to facilitate DER deployment, but also to support the grid. Therefore, VGIC agrees with the SIOGW Report that utilizing DERs to unlock “operational flexibility”, or “the ability of the DSOs [...] to flexibly optimize the use of existing capacity, allowing more rapid connections of DER and loads, while still maintaining grid safety and reliability,”¹ is the highest priority smart inverter operationalization use case. The SIOGW Report discusses the implementation of operational flexibility and specifically configurations for firm import and export limits and non-firm import and export capacity. The Report generally recommends prioritizing implementation of firm and non-firm export limits, including the Limited Generation Profiles (“LGP”) already adopted by the Commission.²

However, given the amount of new flexible load that is expected to come online in California, **VGIC encourages the Commission to begin development of firm and non-firm *import* limits and Limited Load Profiles (“LLP”) as soon as possible.** VGIC believes that the creation of LLPs with firm import limits that vary over a predetermined schedule, which mirror

¹ Report at i.

² Report at iii.

the currently adopted LGPs, should be prioritized strongly by the Commission. A recent motion in this proceeding by Enphase Energy, Inc. (“Enphase”) discussed the importance of advancing LLPs and pursuing their adoption by the utilities as distribution system operators (“DSO”).³ While VGIC and many parties supported the motion, it was denied since “the sub issues Enphase requests to address through the proposed scope expansion are already being addressed in this proceeding.”⁴ The Commission cited many of the issues as being discussed in the SIOWG report, including the development of LLPs and requirements for specific flexible capacity agreements.⁵

In these comments, VGIC focuses on answering the questions outlined in the Ruling specifically with EV use cases in mind.

II. THE COMMISSION SHOULD ESTABLISH STANDARD TERMS AND DEFINITIONS FOR ELEMENTS OF OPERATIONAL FLEXIBILITY.

Stakeholders have been discussing the operational flexibility ideas in the SIOWG for a while in different contexts at the Commission. However, different stakeholders use different names and phrases for the same idea. VGIC has observed discussions around operational flexibility discussed in this High DER Order Instituting Rulemaking (“OIR”), where the SIOWG Report discusses “Operational Flexibility in DER Interconnection Agreements” or “Limited Load Profiles,” but other parties have used phrases including “flexible service connects.” Stakeholders have also brought operational flexibility concepts to the Energization OIR (R.24-01-018), which has discussed “flexible service connections,” “interim service connections,” “bridge service connections,” and “phased energization,” and the Transportation Electrification OIR (R.23-12-

³ *Motion Of Enphase Energy, Inc. to Amend Scoping Ruling and Schedule to Address Optional Flexible Connection Agreements and the Underwriters’ Laboratory 3141 Standard for Power Control Systems in Investor-Owned Utilities’ Energization Rules*

⁴ *Administrative Law Judges’ Ruling Denying Enphase Motion* at 1.

⁵ *Ibid* at 7-8.

008) which has used the phrase “flexible service agreements.” Stakeholders can also use different phrases for technologies and technical capabilities, with even the Commission acknowledging that discussing “smart inverters” is an out-of-date concept since many technologies outside of inverter-based solar and storage can provide the operational flexibility. The SIOGW report therefore discusses “any type of distributed energy resource system that can provide the operational flexibility described in the use cases.”⁶ Several major decisions in the DRIVE OIR (R.18-12-006), including D.20-12-029 and D.22-11-040, refer to Automated Load Management (“ALM”) solutions and EV Energy Management Systems (“EV EMS”) that be used to support operational flexibility. Southern California Edison’s Load Control Management Systems (“LCMS”) Pilot does not use the above terms, rather utilizing the term LCMS to describe both localized, autonomous load control as well as cloud-based, centralized dispatch.

Given the sheer number of phrases and terms being used for operational flexibility, VGIC recommends that the Commission’s Staff Report and subsequent decision on this report include standard terms and definitions for use in *all* proceedings. The current use of terms is confusing and is already leading to stakeholders misunderstanding one another and, at a minimum, creates duplicative discussions about the same topic using different terminologies, which is inefficient for staff, the Commission, and parties. Having standard terms that are used in California to describe elements surrounding these operational flexibility parameters will meaningfully help stakeholders in California better understand and align with each other.

In this set of comments, VGIC will use only the terms and definitions included in the SIOGW Report.

III. RULING QUESTIONS: PRIORITIZATION.

⁶ Ruling at Footnote 1.

1. The SIOWG Report includes prioritization criteria and stakeholder values that are used to inform the prioritization of the business cases and their associated use cases. Prioritization criteria are found in Section 3.4.1 on pages 25-26. Stakeholder values are found in Table 1 of Section 3.4.1 on pages 26-27.

a. Are there any additional prioritization criteria and stakeholder values the Commission could use to determine which business cases to operationalize and what are they and why?

VGIC believes that the Commission should include prioritization criteria that extend beyond technology and business case readiness, which are heavily valued in the Report. Specifically, the Commission should consider how widespread a solution is likely to be used, for example the number of customers that could take advantage of the business case. In “Value to Ratepayer and Society” criteria, the Commission considers progress towards goals that are largely under its own jurisdiction, such as SB 100, but the Commission should also consider how business cases can be used to meet the larger transportation electrification mandates of CARB and statewide electrification goals generally. Business cases that can help California meet these statewide and cross agency mandates and goals should be prioritized.

b. If yes, based on this prioritization criteria and stakeholder values, which business cases and associated use cases are most important for the Commission to prioritize for operation and why?

Based on these additional criteria, VGIC recommends that issues around *import* limits and the development of firm, scheduled LLPs be more heavily prioritized. This would involve increasing prioritization of Use Case 4. Import Limits, particularly scheduled import limits, for Business Case A: Operational Flexibility in DER Interconnection or Limited Load Profiles.

IV. RULING QUESTIONS: CONCEPTS OF FIRM EXPORT LIMITS AND NON-FIRM EXPORT CAPACITY.

2. The following questions are for Business Cases A, B, C, and their associated export related use cases 1-3 in the report.

a. Business Case A - Operational Flexibility in DER Interconnection- Should Distribution System Operators (DSOs) be authorized to set firm and non-firm export limits in the interconnection agreement? What criteria should be used to set the non-firm export limits, such as expected full export capacity or export capacity after eventual grid upgrades?

Yes, generally the DSOs should be able to work with customers to set firm and non-firm export limits in the interconnection agreement. VGIC believes that firm and non-firm export limits can be set both (i) based on temporary or bridge configurations while the distribution system is upgraded or (ii) as an ongoing or permanent solution. Criteria for determining limit values and how often they should change must be well defined.

One case that is likely to emerge is the use of non-firm export capacity as a temporary bridge solution. For example, customers may be willing to pay for distribution upgrades to fully accommodate their DER's export capabilities, but construction of these upgrades may take time. In this case, firm and non-firm export limit values should be revisited once distribution upgrades are completed. However, many customers may also want to use a Limited Generation Profile or another form of non-firm export capacity on an ongoing basis to avoid paying for grid upgrades.

b. Business Case B - Operational Flexibility during Abnormal Conditions- Should DSO be able to utilize schedule and command capability (automation system) to modify firm export limits under abnormal conditions? What criteria should be used for when and how much the firm export limits could be modified, such as expected heat waves or planned grid reconfigurations?

Generally, VGIC cautions against DSOs being able to utilize automated command capabilities that reduce firm export capacity **without explicitly compensating customers for the services provided** by allowing DSOs to access these devices. There are already provisions in the Rule 21 Tariff that allow for automatic disconnections during abnormal conditions to preserve grid safety, but Business Case B discusses additional abilities for DSOs to either communicate additional abnormal or forecasted abnormal conditions or to directly control devices. While VGIC

understands that this business case may be beneficial for DSOs, when a customer signs an interconnection agreement with a firm export capacity, they expect to be able to export to that limit across the year. Customers typically receive compensation for exports under tariffs such as the Net Billing Tariff (“NBT”), other feed-in-tariffs, or demand response programs. The utilities are currently working on V2X pilots and the Commission is working on developing a technology neutral export rate option in the Advanced Demand Flexibility OIR. In order to allow the DSOs to utilize command capabilities to reduce firm export limits, customers must be compensated for their forgone revenue or service. Otherwise, customers are unlikely to agree to such terms and may forgo providing exports to the grid at all. VGIC recommends policy discussions surrounding proper compensation for customers forgoing exports occur and conclude *before* the Commission implements Business Case B, particularly if DSOs are able to directly limit exports.

c. Business Case C - Operational Flexibility for Distribution Services under Normal Conditions- Should DSOs be authorized to dynamically modify non-firm export limits using schedule and/or command capability (automation system) to DER facilities on an ongoing basis?

If customers are guaranteed access to a minimum firm export limit, then VGIC supports dynamic signals to encourage DER operations in support of the grid. VGIC does not support DSOs using automatic command capability to control devices on a regular basis, but VGIC does support Business Case C as explained in the report, “DER operators may *voluntarily* take actions in response to incentives under normal conditions, or even [...] pending abnormal conditions. [*emphasis added*].”⁷ DSOs may want to increase non-firm export limits during times of high load, such as heatwaves, where distribution capacity exists to accommodate increased exports. For these scenarios, VGIC agrees that communication protocols should be developed to allow for this

⁷ Report at 46.

capability. In these scenarios, customers are likely to be incentivized to provide these exports for additional compensation under tariffs or demand response programs. The Commission should also consider whether additional or different forms of compensation are appropriate when different grid services are being provided (i.e., an unbundled “value stack”).

d. Business Case A - Operational Flexibility in DER Interconnection- Should DSOs be authorized to set minimum export requirements in the interconnection agreement? What criteria (time, duration, amount) should be included?

VGIC answers this question for vehicle-to-grid (“V2G”) exporting use cases. For V2G applications, VGIC does not recommend setting minimum export requirements at this time. EVs are purchased, as all vehicles are, for transportation. While grid services can be provided by EVs, transportation capabilities will be prioritized by customers. This means that EVs will not always be plugged in and it would be unreasonable to have minimum export requirements in V2G interconnection agreements at this time.

e. Business Case B - Operational Flexibility during Abnormal Conditions- Should DSOs be able to utilize schedule and command capability (automation system) to modify minimum export requirements under abnormal conditions?

For V2G applications, VGIC does not recommend utilizing command capabilities to instate or modify minimum export requirements at this time. As discussed above, serving the grid is not the primary role of EVs and transportation capabilities may need to be prioritized. For example, if abnormal grid conditions that require additional exports are expected to continue for extended periods of time, EVs may have to hold a charge in order to drive later. Establishing or increasing a minimum export requirement could worsen the grid condition by requiring the EV to recharge at a later time in order to drive.

f. Business Case C - Operational Flexibility for Distribution Services under Normal Conditions- Should DSOs be authorized to dynamically modify minimum export requirements using schedule and/or command capability (automation system) to DER facilities on an ongoing basis?

Given that VGIC generally opposes creating minimum export requirements for V2G applications, VGIC has no additional comment on this question.

g. If yes to the questions above, what should be the scheduling structure (e.g., Comma Separated Value (CSV)), interaction (e.g., acknowledgment and error handling), and granularity level (e.g., monthly, weekly, daily, hourly, or others)?

N/A.

3. Under what conditions, if any, should an interconnection agreement be updated to modify firm export limits and/or non-firm export capacity?

a. For example, following a distribution circuit capacity upgrade, permanent circuit reconfiguration, or when the distribution system's capacity decreases to a level where non-firm export capacity is no longer available.

Generally, VGIC does not believe that firm export limits should be reduced once a customer has signed an interconnection agreement for a specific export capability. As discussed above, if a DSO wants to reduce a customer's ability to export below the minimum amount agreed to in their signed interconnection agreement, compensation for any forgone revenue from those exports should be provided. If a distribution circuit is upgraded and can accommodate additional exports, then the interconnection agreement could be updated to allow for additional exports beyond current limits. The Commission should establish rules and regulations detailing how the DSOs would approach customers to increase export limits and how multiple customers with limited export capacities on the same circuit would be approached in order to ensure fair access to increased capacity. VGIC does not have specific recommendations on how DSOs should fairly allocate increased capacity at this time and looks forward to further collaboration with stakeholders on the topic.

4. Should firm and non-firm export agreements be extended to DER owners with existing interconnection agreements? If so, how?

VGIC has no comment at this time.

5. The following questions are for Business Cases A, B, C, and their associated import related Use Case 4 in the report. A Business Case and Use Case matrix is available in Appendix A.

a. Business Case A - Operational Flexibility in DER Limited Load Profiles - Should Distribution System Operators (DSOs) be authorized to set firm and non-firm import limits in the connection agreement? What criteria should be used to set the non-firm import limits, such as overall import needs even if some time periods are constrained by firm import limits, or the expected import capacity after eventual grid upgrades?

Overall, VGIC believes that the Commission should prioritize the establishment of Limited Load Profiles (“LLPs”), especially LLPs that create a firm import schedule that varies over different hours or seasons. The use of LLPs has the potential to substantially increase transportation electrification in California by allowing customers to take advantage of existing distribution capacity. As distribution capacity generally increases on the grid, import limits can be modified to allow for additional imports.

The Report recommends focusing on the implementation of the Limited Generation Profiles (“LGP”) to allow for scheduled exports. Currently, the LGPs include options for customers to choose different firm operational schedules. VGIC recommends that parallel LLPs also be developed and implemented as soon as possible to allow for customers to schedule their imports. In cases where 24/7 available capacity is less than the planned load for a customer’s site, the DSO could study the system under different LLP configurations (i.e., “24- hourly,” “Block,” and “18-23-fixed”, or other simpler schedules as discussed below) and provide these options to the customer, who could then decide which profile works best for them.

VGIC also supports the development of non-firm capacity agreements, but as discussed in the report, communications and other protocols for communicating non-firm capacity will involve

“testing, and deployment over an estimated 2 to 10 years with pilot projects and a focus on the larger DER facilities.”⁸ Therefore, the Commission should focus on creating a varying schedule of firm import limits to begin, leveraging the work that has already been done for the LGPs to develop this program on the load side.

Customers may agree to a permanent LLP import schedule, but VGIC anticipates that many customers will want, as a first choice, the distribution system to be upgraded to allow for the site to be fully energized at all times. In a “bridge” scenario, the customer would only agree to a particular limit during construction, with a new energization agreement being signed after upgrades are completed. VGIC recommends that the Commission create a temporary LLP option with this scenario in mind.

b. Business Case B - Operational Flexibility during Abnormal Conditions, Should DSOs be able to utilize schedule and command capability (automation) to modify firm import limits under abnormal conditions? What criteria should be used for when and how much the firm import limits could be modified, such as expected heat waves or planned grid reconfigurations?

VGIC cautions against utilizing DSO command capabilities to limit electricity imports below *firm* import limits established in an interconnection agreement, especially for EVs at this time. As discussed above, EVs are primarily used for transportation, and customers may need to charge their vehicles during times of grid stress. VGIC fully supports the Commission’s efforts to send economic signals to shape charging, including the development of dynamic pricing that would allow customers to see more of the real-time costs of their energy. However, the utilities maintain their obligation to serve customers, and changing a customer’s import capability without significant amounts of notice would severely diminish customer confidence in EVs generally.

⁸ Report at iv.

Therefore, VGIC does not believe that the DSOs should be able to modify firm import limits using command capabilities at this time.

Business Case B is distinct from the development of the firm import LLPs discussed above because establishing a set profile in the interconnection agreement allows customers to plan their imports around pre-defined limits that are agreed to in advance. Additionally, real-time changes to *non-firm* import limits could be developed, but the Commission will have to more thoroughly flesh out how much advance notice customers would get, how a DSO would determine how much to reduce the import limits of individual customers on a circuit, etc. Therefore, VGIC supports continuing discussions on this use case for imports, but the development of *firm LLPs* should be prioritized given the opportunity to leverage recent LGP development work.

c. Business Case C - Operational Flexibility for Distribution Services under Normal Conditions- Should DSOs be authorized to dynamically modify non-firm import limits using schedule and/or command capability (automation system) to DER facilities on an ongoing basis?

Similar to our above answer for exports, VGIC considers Business Case C to discuss *voluntary* actions, in this case, to reduce imports. Generally, VGIC supports sending economic signals to increase or decrease imports, such as through the development of demand response programs, managed charging programs, or dynamic rates. VGIC does not believe that DSOs should be able to use command capability to reduce import limits on an ongoing basis without compensation commensurate with the values provided to the DSO.

d. Business Case A - Operational Flexibility in DER Limited Load Profiles - Should DSOs be authorized to set minimum import requirements in the interconnection agreement? What criteria (time, duration, amount) should be included?

For EV applications, VGIC does not recommend setting minimum import requirements at this time. As discussed above, EVs will have to prioritize transportation capabilities and services and will not always be charging or discharging. This may mean that there would be times where no EVs are stationed and charging, making minimum import requirements difficult to meet.

e. Business Case B - Operational Flexibility during Abnormal Conditions- Should DSOs be able to utilize schedule and command capability (automation system) to modify minimum import requirements under abnormal conditions?

Since VGIC generally does not recommend establishing minimum import requirements, we do not have additional comments on this question.

f. Business Case C - Operational Flexibility for Distribution Services under Normal Conditions- Should DSOs be authorized to dynamically modify minimum import requirements using schedule and/or command capability (automation system) to DER facilities on an ongoing basis?

Since VGIC generally does not recommend establishing minimum import requirements, we do not have additional comments on this question.

g. If yes to the questions above, what should be the scheduling structure (e.g., CSV), interaction (e.g., acknowledgment and error handling), and granularity level (e.g., monthly, weekly, daily, hourly, or others)?

Generally, VGIC believes that firm, scheduled LLPs should be implemented by the Commission before more dynamic scheduling of non-firm capacity is implemented. VGIC does not have specific recommendations for the scheduling structure at this time.

6. Regarding the granularity of hourly scheduling values for Power Import Limiting profiles:

a. Do you view 288 hourly values spread across three possible configurations (“24-hourly,” “Block,” and “18-23-fixed”), as described in Resolution E-5296 regarding Limited Generation Profiles, as an adequate symmetrical construct on the load side?

VGIC supports the use of the three scheduling profiles developed for the LGPs to develop Limited Load Profiles (“LLPs”), but we recommend looking at simpler schedules to use as well.

The use of 24 unique values under the three LGP configurations will provide customers with different options to consider and will help to maximize the amount of load that can be energized. However, for load, customers may prefer to implement a simpler schedule that is easier to manage and maintain.

b. If not, do you believe a different number of hours and structure(s) should be used for scheduling Power Import Limiting profiles?

One structure that VGIC recommends considering for scheduling LLPs is a 4-value structure with firm import limits that vary by Winter versus Summer and Peak versus Off-peak hours. A 12-value monthly LLP, while inappropriate to maximize value to the grid for exports in the LGP, may also be suitable for LLPs, as it's an easier paradigm to understand and plan around.

7. Under what conditions, if any, should a Service Agreement be updated to modify firm import limits and/or non-firm import capacity?

a. For example, following a distribution circuit capacity upgrade, permanent circuit reconfiguration, or when the distribution system's capacity decreases to a level where non-firm import capacity is no longer available.

The Commission should establish an option for LLPs to be used as a temporary profile that can later be updated once distribution upgrades are constructed to accommodate the full load of that customer. This scenario is already emerging in California on an ad hoc basis and can be scaled immediately to all interested customers. Other conditions, such as the allocation of capacity from shared distribution system upgrades that are built years after a customer signed their energization agreement, require further discussion.

8. The report uses the term "Limited Load Profiles" to describe the agreement for energization. Are there any suggested alternative terms to describe providing energy and service agreements to customers?

VGIC does believe that LLPs as a mirror of the LGPs - specific interconnection agreements that provide options of schedules that can be followed with different import limits across the year. The Report discusses more direct responses to real-time grid conditions in Business Cases B and C that operate to respond to real-time grid conditions. Since those Business Cases may either be distinct or layered on top of an LLP profile, those use cases may probably have a distinct name that is separate from LGP and LLP. VGIC also reiterates its recommendation that the Commission clearly define the various terms related to these concepts, as detailed in Section II above.

V. RULING QUESTIONS: EV (BUSINESS CASE E).

10. Regarding any initial efforts to deploy EV charging projects through flexible connection agreements:

- a. How many EV charging sites (including capacity), that are either operational or under development, are piloting or using a form of flexible connection agreement that relies on power import limiting schedules? i. Note: If responses include confidential information, please include a redacted and non-redacted version.*
- b. For sites as described in 12(a), how is PG&E, SCE, and SDG&E differentiating between firm and non-firm levels of energization and / or export from a contractual and operational standpoint?*
- c. What is the preferred temporal granularity for power import limiting and/or power export limiting schedules?*
- d. What communication pathways or protocols, if any, are required for the DSOs to communicate with and send commands to the EVSE at EV charging sites?*

VGIC has no comment at this time but may provide additional input in reply comments.

VI. RULING QUESTIONS: REGULATORY ISSUES.

11. Which regulatory proceedings (existing or new) should address these recommended capabilities for generation limiting with non-firm export capacity? (Note: The IOUs' Rule 21 tariffs currently address firm export limits).

VGIC is agnostic to the regulatory venue to address non-firm import and export capacity. Overall, VGIC believes that developing these capabilities in this High DER OIR is most efficient. At the same time, for exporting resources, this High DER proceeding will have to coordinate with

the Rule 21 proceeding since export interconnection agreements are generally governed by Rule 21 and, once modified, must be implemented in Rule 21.

12. Which regulatory proceedings (existing or new) should address these recommended capabilities for load limiting with firm import limits and non-firm import capacity?

VGIC is overall agnostic to the regulatory venue to address LLPs; however, we emphasize that the Commission should take this issue up quickly. VGIC believes that this High DER proceeding is likely the appropriate venue to develop the LLPs and general flexible service connection offerings, given the Report and recent High DER Staff Proposal. On the other hand, the Commission has explicitly stated that the energization rulemaking should take up issues surrounding customer energization.⁹ As highlighted by VGIC in both proceedings, the High DER proceeding and the Energization proceeding will likely need to be closely coordinated, as once the LLPs are developed, energization rules will need to be modified to include them.

13. What regulatory requirements would need to be defined to provide operational guidelines on the fair allocation of non-firm capacity to different DER facilities? What issues would need to be resolved?

VGIC has no comments on fair allocation of capacity at this time.

14. How should regulations and tariffs for export and import limits be designed for DERs that have both load and generation attributes, such as stand-alone battery storage, PVpaired storage, and bidirectional EV charging? Should power import and export issues be addressed in separate rulemakings working in parallel, or should they be combined into a single rulemaking?

Overall, VGIC believes that it is most efficient to develop standard tariffs or general program parameters for export and import limits in this High DER OIR. These can then be taken to different proceedings, including Rule 21 and the Energization OIR, to be implemented in the required rules (i.e., Rules 2, 15, 16, 21, and 29/45). VGIC does not believe that technologies need

⁹ ADMINISTRATIVE LAW JUDGES' RULING DENYING ENPHASE MOTION at 1.
<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M534/K105/534105698.PDF>

to be discussed separately, but some nuances may need to be considered for individual technologies and the use cases of those technologies (i.e., EV's primary requirements to provide transportation services). Those nuances can also be discussed during the implementation of a tariff change in Rule 29/45, which is EV-specific.

VGIC believes there is value in coordinating discussions for both imports and exports in one proceeding. This ensures that technologies that both import and export, such as storage and bidirectional EVs, can have both limits directly discussed. Additionally, as shown in the report, many of the technical requirements like power control systems, communications protocols, safety standards, and DER management systems ("DERMS") are the same for import and export limits. Therefore, it makes sense to discuss both in the same proceeding. Once a program or tariff structure is developed, discussions may have to be separated into the Rule 21 and Energization proceedings.

15. Annex B briefly addresses the handling of unused capacity. What issues do you see related to managing customer unused capacity? What, if anything, should the Commission do relative to unused capacity? Should this be addressed in the IOUs' tariffs?

While the issue of unused or "stranded" capacity is worth considering, it is unclear that there is significant unused capacity on the grid today or that this will emerge as an important issue in the future. The Report does not highlight unused capacity as a prevalent problem for California today. VGIC recommends that additional study be done on the most common ways unused capacity emerges and, specifically, if certain customer types or use cases lead to unused capacity more often than others. Given the importance of expanding the distribution grid and the additional load that is expected throughout California, the Commission should spend its resources addressing the more well-defined and beneficial use cases, such as LGP and LLP, described above. In future venues, unused capacity can be studied and discussed further.

16. What existing standards (such as technical, safety, testing, and operational) should the Commission consider referencing in developing rules and regulations for Business Cases and Use Cases in the report? What new standards will be needed to inform CPUC regulation and policy?

VGIC has no comment at this time but may provide additional input in reply comments.

VII. RULING QUESTIONS: CYBERSECURITY REQUIREMENTS.

19. Do you agree with the Commission establishing Paths F, G, and H based on SIO-CS prioritization? If yes, why? If not, why not?

20. Questions related to Path F (Recommend that DSOs use the SIO-CS Phase 1 Primary DER Cybersecurity Requirements in the short term as part of their cybersecurity requirements for DER facilities):

21. Questions related to Path G (Initiate the development of Phase 2 DER Cybersecurity Requirements based on SIO-CS Phase 1):

22. Questions related to Path H (Endorse the establishment of testing and certification programs):

VGIC has no comment at this time but may provide additional input in reply comments.

VIII. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the Working Group Reports. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Interim Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

Date: July 8, 2024